

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2185

Barton 138 kV
October 27, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2185. The Interconnection Customer proposes a 720.8 MW natural gas combined cycle facility interconnecting at the Barton 138 kV substation in the Pennsylvania area. The request was received on September 22, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 437 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,281,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Kramer-Barton 138 KV, the Barton-Weld 138 KV loads to 210.4% of its 236 MVA rating. Project DFAX: 72.1%.

For the loss of Barton-Weld 138 KV Line, the Kramer-Brton 138 KV line loads to 194.9% of its 252 MVA rating. Project DFAX: 64.7%.

Kramer-Island Road 138 KV: 193.8% loading (249 MVA) following the outage of the Weld-Barton 138 KV; DFAX = 64.7%.

Lakemont-Weld 138 KV: 149.5% loading (185 MVA) following the outage of the Kramer-Barton 138 KV Line; DFAX = 37.4%.

Menaul Tap-Island Road 138 KV line: 108.9% loading (215 MVA) following the outage of the Island Rd-Subestacion Escuela li Substation 138 KV L; DFAX = 29.2%.

The Scuykill-Tunnel 138 KV L is overloaded to 108.2% of its 259 MVA emergency rating for the loss of the Island Rd-Subestacion Escuela li 138 KV. The project contributes 8.7% DFAX (62.4 MW) to this overload.

For the loss of Barbadoes-Plymouth Meeting 138 KV, the Granby-Wauzeka 138 KV loads to 104.2% of its 263 MVA rating. Project DFAX: 7.1%.

The Island Road-Subestacion Escuela li 138 KV is overloaded to 103.5% of its 257 MVA emergency rating for the loss of the Yreka-Tunnel 138 KV. The project contributes 12.6% DFAX (91.0 MW) to this overload.

For the loss of Weld-Barton 138 KV, the Darbyville-W Deptford Energy Sta 138 KV line loads to 102.5% of its 211 MVA rating. Project DFAX: 35.3%.

The Angora-Maplewood 138 KV Ckt 1 is overloaded to 100.8% of its 290 MVA emergency rating for the loss of the Angora-Maplewood 138 KV Circuit 2. The project contributes 12.5% DFAX (90.3 MW) to this overload.

Angora-Maplewood 138 KV #2: 100.8% loading (290 MVA) following the outage of the Maplewood-Angora 138 KV #1; DFAX = 12.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Barton-Weld 138 KV	Rebuild line with higher-capacity conductor	\$1,914,000	100% (\$1,914,000)
Kramer-Brton 138 KV line	Rebuild line with higher-capacity conductor	\$3,206,000	100% (\$3,206,000)
Kramer-Island Road 138 KV	Rebuild line with higher-capacity conductor	\$547,000	100% (\$547,000)
Lakemont-Weld 138 KV	Rebuild line with higher-capacity conductor	\$1,402,000	100% (\$1,402,000)
Menaul Tap-Island Road 138 KV line	Reconductor line	\$513,000	100% (\$513,000)
Scuykill-Tunnel 138 KV L	Reconductor line	\$506,000	86% (\$436,000)
Wauzeka-Granby 138 KV	Reconductor line	\$527,000	60% (\$316,000)
Island Road-Subestacion Escuela li 138 KV	Reconductor line	\$766,000	78% (\$594,000)
Darbyville-W Deptford Energy Station 138 KV L	Reconductor line	\$409,000	100% (\$409,000)
Angora-Maplewood 138 KV Cir. 1	Reconductor line	\$528,000	68% (\$357,000)
Angora Substation-Maplewood 138 KV Line 2	Reconductor line	\$867,000	68% (\$587,000)
	Total allocated to project		\$10,281,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.